

HEALTHCARE DATA & M&E

# Syphilis/VDRL Test Positivity Rate

## Indicator Definition & Monitoring Playbook

*Portfolio Demonstration Project*

### PROJECT PURPOSE

A simulated professional work sample demonstrating indicator definition, measurement logic, health data quality assurance, monitoring, interpretation and reporting.

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## 1. Executive Summary

This playbook provides a standardized approach for defining, measuring, validating, interpreting and reporting the Syphilis/VDRL Test Positivity Rate. It is developed as a portfolio demonstration using a realistic public-health scenario: concerns about a possible increase in syphilis detection.

The project deliberately does not assume or claim that a nationwide increase has been established. Instead, it demonstrates how routine testing data could be used to monitor test positivity and identify patterns requiring further review.

### KEY PRINCIPLE

An observed change in test positivity is a signal for investigation. It is not, by itself, proof of a change in national syphilis incidence or prevalence.

## 2. Background & Context

Syphilis surveillance depends on reliable identification of testing patterns, laboratory results and the populations represented in routine health-service data. A reported increase in positive results may reflect several factors, including increased testing, changes in the population tested, changes in testing practices, improved reporting or a true change in infection patterns.

For this reason, a standardized indicator is useful. The proposed indicator focuses on the proportion of valid syphilis/VDRL tests that are positive during a defined reporting period.

## 3. Indicator Definition

| Field                   | Definition                                                                                |
|-------------------------|-------------------------------------------------------------------------------------------|
| Indicator name          | Syphilis/VDRL Test Positivity Rate                                                        |
| Indicator type          | Percentage / proportion                                                                   |
| Unit of measure         | Percent (%)                                                                               |
| Purpose                 | Monitor the proportion of reported valid syphilis/VDRL tests that are positive over time. |
| Recommended aggregation | Monthly                                                                                   |
| Core population         | Individuals represented in the available valid testing data                               |

### 3.1 Indicator Formula

**Syphilis/VDRL Test Positivity Rate (%) =**

**(Number of valid positive tests ÷ Number of valid tests performed) × 100**

### 3.2 Numerator

Number of valid syphilis/VDRL tests with a positive or reactive result during the reporting period, according to the result categories defined for the dataset.

### 3.3 Denominator

Total number of valid syphilis/VDRL tests performed during the same reporting period and having an interpretable result.

## 4. Indicator Measurement Rules

### 4.1 Inclusion in the numerator

- The test was performed within the reporting period.
- The record identifies the relevant syphilis/VDRL test.
- The result is recorded as positive/reactive under the defined result convention.
- The record is not a confirmed duplicate.
- The record contains sufficient information to establish that it is a valid test result.

### 4.2 Inclusion in the denominator

- The test was performed within the reporting period.
- The test is a valid syphilis/VDRL test.
- The result is interpretable.
- The record is not a confirmed duplicate.

### 4.3 Exclusion criteria

- Duplicate records confirmed through available identifiers.
- Cancelled or unperformed tests.
- Records with no interpretable result.
- Clearly invalid or erroneous records.
- Tests outside the reporting period.

### 4.4 Missing results

#### DATA QUALITY RULE

A missing test result must not automatically be classified as negative. Missing or invalid results should be identified and handled according to documented data-quality procedures.

### 4.5 Repeat tests

For the primary indicator, the unit of measurement is the valid test episode rather than the unique individual. A person may therefore contribute more than one valid test over time. Repeat testing should be documented as a limitation when interpreting trends.

### 4.6 Laboratory interpretation

#### CLINICAL/LABORATORY CAUTION

A reactive VDRL result should not automatically be interpreted as confirmed active syphilis. This indicator measures test positivity, not confirmed disease diagnosis.

## 5. Data Elements Required

### 5.1 Essential data elements

| Data element | Why it is needed                            |
|--------------|---------------------------------------------|
| Test date    | Determines the reporting period.            |
| Test type    | Identifies the relevant syphilis/VDRL test. |

|             |                                                           |
|-------------|-----------------------------------------------------------|
| Test result | Determines positive/reactive versus non-positive results. |
| Facility    | Supports facility-level analysis.                         |
| County      | Supports geographic analysis.                             |

## 5.2 Supporting data elements

| Data element             | Why it is useful                                     |
|--------------------------|------------------------------------------------------|
| Patient/record ID        | Supports duplicate review and longitudinal analysis. |
| Age or date of birth     | Enables age-group disaggregation.                    |
| Sex                      | Enables sex-disaggregated analysis.                  |
| Pregnancy status         | Identifies pregnancy-related testing.                |
| Service point            | Distinguishes ANC, PNC, OPD, STI and other settings. |
| ANC visit/status         | Supports ANC-specific analysis.                      |
| Gestational age          | Adds pregnancy-related context where available.      |
| Specimen collection date | Supports timeliness and laboratory workflow review.  |
| Result/reporting date    | Supports reporting timeliness checks.                |

## 5.3 Potential disaggregation

| Category                   | Examples                                                                               |
|----------------------------|----------------------------------------------------------------------------------------|
| Demographic                | Age group; sex                                                                         |
| Geographic                 | County; sub-county; facility                                                           |
| Service/population context | ANC; PNC; OPD; STI/sexual health services; other testing settings                      |
| Pregnancy context          | Pregnant women; ANC first visit; subsequent ANC visit; gestational age where available |
| Time                       | Month; quarter; year                                                                   |

## 6. Data Sources & Collection Method

| Potential source            | Information that may be obtained                               |
|-----------------------------|----------------------------------------------------------------|
| Laboratory register / LIMS  | Test date, test type, result and specimen-related information. |
| EMR/EHR                     | Demographics, pregnancy status and service context.            |
| ANC/PNC registers           | Maternal service and pregnancy-related information.            |
| STI/sexual health registers | Syphilis testing and relevant clinical context.                |
| Routine reporting system    | Aggregated facility/county reporting.                          |

### 6.1 Primary source vs reporting source

The primary source is where the information is originally recorded, such as a laboratory register or LIMS. The reporting source is where data are aggregated and submitted for monitoring. Indicator review should, where possible, trace reported values back to their source records.

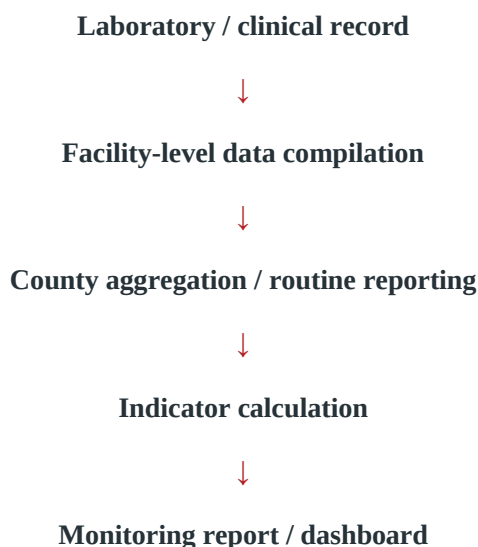
### 6.2 Recommended frequency

For this portfolio example, data are assumed to be collected routinely, aggregated monthly and reviewed monthly. Quarterly or annual aggregation may be appropriate for higher-level trend assessment.

## 7. Conceptual Data Flow

Patient receives testing





## 8. Data Quality Checks

| Dimension      | Checks                                                                                     |
|----------------|--------------------------------------------------------------------------------------------|
| Completeness   | Required fields populated: date, test type, result, facility and county.                   |
| Validity       | Dates, ages, test types and result categories are within expected values.                  |
| Uniqueness     | Potential duplicates identified using available patient/test/date/facility identifiers.    |
| Consistency    | Facility and county relationships and related fields agree.                                |
| Timeliness     | Specimen/result/reporting dates support timely inclusion in the intended reporting period. |
| Missingness    | Missing results are identified and not automatically converted to negative results.        |
| Outlier review | Unexpected changes in testing volume or positivity are investigated before interpretation. |

## 9. Indicator Interpretation

### INTERPRETATION PRINCIPLE

A change in the indicator is a signal for investigation and not automatically proof of a change in disease burden.

### 9.1 Scenario-based interpretation

| Observed pattern                                              | Possible interpretation                                                             | Recommended action                                                                                 |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Positivity increases while testing volume is stable           | A higher proportion of reported tests are positive.                                 | Validate data; review the trend over multiple periods; examine geographic and population patterns. |
| Positivity increases while testing volume rises substantially | More positive tests may partly reflect increased testing.                           | Review changes in screening, facility reporting, campaigns and testing practices.                  |
| One facility shows an unusual spike                           | Could reflect a local signal, different case mix, referral patterns or data issues. | Conduct facility-level data-quality and context review before labeling a hotspot.                  |
| ANC positivity increases                                      | A higher proportion of reported ANC tests are positive.                             | Review testing volume, ANC population, data quality and                                            |

|                                                            |                                            |                                                                                       |
|------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------|
|                                                            |                                            | treatment/follow-up information where available.                                      |
| Positivity increases consistently across multiple settings | A stronger surveillance signal is present. | Consider further epidemiological or programmatic investigation after data validation. |

## 9.2 What the indicator cannot tell us

- National population prevalence of syphilis.
- Incidence of new syphilis infections.
- The number of unique infected individuals.
- Whether a positive VDRL result represents active infection.
- The cause of an observed increase.
- Whether transmission has increased.

## 9.3 Decision framework

**OBSERVE → VALIDATE → COMPARE → INVESTIGATE → INTERPRET → ACT**

Use this sequence to prevent premature conclusions from routine testing data.

# 10. Reporting & Visualization

## 10.1 Core monthly reporting table

| Reporting period | Tests performed | Positive tests | Positivity rate |
|------------------|-----------------|----------------|-----------------|
| January          | 5,000           | 75             | 1.5%            |
| February         | 5,200           | 83             | 1.6%            |
| March            | 5,100           | 102            | 2.0%            |
| April            | 5,300           | 133            | 2.5%            |

The numerator and denominator should be reported alongside the percentage so that users can assess the context behind the indicator.

## 10.2 Recommended visualizations

- Line chart: monthly positivity rate.
- Line or column chart: monthly testing volume.
- County/facility table or map: positivity with testing volume shown.
- Age and sex disaggregation charts.
- ANC/PNC or service-context comparison where available.
- Data-quality panel: completeness, missing results, duplicates and reporting timeliness.

## 10.3 Recommended dashboard layout

| Section                   | Content                                                             |
|---------------------------|---------------------------------------------------------------------|
| KPI cards                 | Total tests; positive tests; positivity rate; reporting facilities. |
| Trend                     | Monthly positivity and testing volume.                              |
| Geography                 | County/facility positivity with context on testing volume.          |
| Demographics              | Age and sex patterns.                                               |
| Pregnancy/service context | ANC/PNC and other service settings where available.                 |
| Data quality              | Completeness, messiness, duplicates and reporting anomalies.        |

## 10.4 Example reporting narrative

### EXAMPLE

Syphilis test positivity increased from 1.8% in March to 2.3% in April among reported valid tests. Testing volume remained relatively stable during the period. The increase should be reviewed alongside facility-level reporting, testing patterns and data-quality indicators before conclusions are drawn regarding changes in disease burden.

## 11. Roles & Responsibilities

| Illustrative role           | Key responsibility                                                    |
|-----------------------------|-----------------------------------------------------------------------|
| Laboratory staff            | Perform/report testing and ensure accurate recording of results.      |
| Data/M&E officer            | Compile, clean, validate and calculate the indicator.                 |
| Facility in-charge          | Review facility results and address data-quality issues.              |
| County M&E/health team      | Aggregate facility data, review trends and follow up anomalies.       |
| Program manager             | Interpret findings and coordinate programmatic review or response.    |
| Data/quality assurance team | Conduct periodic data-quality assessments and verify reported values. |

These roles are illustrative for this portfolio project and should not be interpreted as an official Kenyan reporting structure.

## 12. Common Errors & Troubleshooting

| Common issue                                     | Recommended response                                                                                  |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Missing result                                   | Do not classify as negative; trace the source record and apply the documented missing-data rule.      |
| Duplicate record                                 | Review available identifiers and remove only confirmed duplicates.                                    |
| Inconsistent result terminology                  | Standardize valid categories during data cleaning.                                                    |
| Sudden testing-volume spike                      | Check new reporting facilities, campaigns, backlog entry, duplicates or workflow changes.             |
| Positive cases increase but positivity is stable | Assess whether increased testing volume explains the increase in positives.                           |
| Facility reports unusually high positivity       | Review data quality, testing volume, case mix, referrals and testing practices before interpretation. |

## 13. Worked Example

Assume a facility reports 5,000 valid VDRL tests in April, of which 125 are positive/reactive.

$$\text{Positivity rate} = (125 \div 5,000) \times 100 = 2.5\%$$

Therefore, the April Syphilis/VDRL Test Positivity Rate is 2.5%.

If March was 1.8%, the change is 0.7 percentage points. The relative increase from 1.8% to 2.5% is approximately 38.9%. These are different measures and should not be described interchangeably.

## 14. Quick Reference Indicator Sheet

| Field                 | Quick reference                                                                          |
|-----------------------|------------------------------------------------------------------------------------------|
| Indicator             | Syphilis/VDRL Test Positivity Rate                                                       |
| Purpose               | Monitor the proportion of reported valid tests that are positive.                        |
| Numerator             | Valid positive/reactive tests.                                                           |
| Denominator           | Valid syphilis/VDRL tests with interpretable results.                                    |
| Formula               | $(\text{Positive tests} \div \text{valid tests}) \times 100$                             |
| Unit                  | Percent (%)                                                                              |
| Recommended frequency | Monthly                                                                                  |
| Core disaggregation   | Age, sex, county, facility, service context, pregnancy/ANC status where available.       |
| Primary sources       | Laboratory register/LIMS, EMR/EHR, ANC/PNC and relevant clinical registers.              |
| Key caution           | Test positivity is not equivalent to population prevalence or confirmed active syphilis. |

## 15. Limitations

- Routine testing data may not represent the entire population.
- Testing practices and screening policies may change over time.
- Repeat testing may influence the positivity rate.
- Missing or incomplete data may bias the indicator.
- VDRL is a non-treponemal test and requires appropriate clinical/diagnostic interpretation.
- A rise in positivity does not independently establish a rise in national syphilis incidence or prevalence.
- County and facility comparisons may be affected by differences in populations served, referral patterns and testing practices.

## 16. Portfolio Disclaimer

### PORTFOLIO DEMONSTRATION PROJECT

This indicator playbook is a simulated professional work sample developed to demonstrate skills in healthcare indicator definition, monitoring and evaluation, data quality management and analytical interpretation. It is not an official national surveillance guideline and does not represent the policies, recommendations or official position of the Government of Kenya, Ministry of Health, WHO, or any other organization.

## 17. References & Evidence Base

Before publication, the final portfolio version should include verified, authoritative references supporting the laboratory interpretation of VDRL, syphilis testing algorithms, maternal/ANC syphilis screening guidance and any Kenya-specific surveillance context used in the background.

- World Health Organization (WHO) relevant syphilis testing and surveillance guidance.



- Kenya Ministry of Health relevant national STI/syphilis and maternal health guidance, where applicable.
- Peer-reviewed literature and recent Kenyan evidence used to frame the possible surveillance concern.

**Daisy Wambui Mwaura | Healthcare Data & M&E**